

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

18th May 2023

Subject CB1 – Business Finance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Which one of the following is the goal of the Financial Manager?
- A.** To invest in projects if the rate of return is greater than the cost of borrowing.
 - B.** To invest in projects that display a positive net present value.
 - C.** To determine a range of social objectives beyond making profits.
 - D.** To focus attention on the impact of company's operation on climate change. **[2]**
- Q. 2)** Internal growth is preferred by firms that wish to:
- A.** avoid unnecessary Government intervention.
 - B.** expand geographically.
 - C.** to share the financial burden and the risk of the project.
 - D.** to acquire assets or experience. **[2]**
- Q. 3)** Which of the following statement is 'true' for a Pre-payment crowdfunding?
- A.** Investors receive interest on the money they lend, and their capital is repaid over time.
 - B.** Investors buy shares and benefit if the business succeeds.
 - C.** In return for giving money, participants are rewarded by receiving a service or a product.
 - D.** There are no financial rewards and the donors derive satisfaction from helping charitable causes. **[2]**
- Q. 4)** Which of the following statement is not true in case of limited companies?
- A.** If shares have been issued 'partly paid' then in the event of liquidation, shareholders will only be liable to pay the outstanding installments.
 - B.** If the shares have been issued on 'partly paid' basis, at a premium to their par value, only a part of this 'share premium' is payable at the outset.
 - C.** If the company becomes insolvent, creditors cannot claim further payment from the shareholders personal wealth beyond the fully paid value of their shares.
 - D.** Article of Association set out the internal rules for running the company. **[2]**
- Q. 5)** Which of the following correctly describes operating leases?
- I.** The owner of the asset will retain most of the risks associated with owning the asset.
 - II.** The leaser takes on most of the risks associated with owning the asset.
 - III.** The lease will be for a period similar to the likely life of the asset.
 - IV.** The lease will be for a period substantially shorter than the likely life of the asset.
- A.** Only I and III are correct.
 - B.** Only II and III are correct.
 - C.** Only II and IV are correct.
 - D.** Only I and IV are correct. **[2]**
- Q. 6)** Which of the following is NOT a correct interpretation of the prudence concept?
- A.** An asset that cost ₹3,00,000 was professionally revalued at ₹4,00,000 and that valuation has been recognised in the financial statements at the revised value.

- B. An asset that could be sold between ₹5,00,000 and ₹7,00,000 has been valued at ₹7,00,000 in the financial statements.
- C. An asset that could be sold between ₹5,00,000 and ₹6,00,000 has been valued at ₹4,50,000 in the financial statements.
- D. An intangible asset could be worth up to ₹5,00,000, but the asset's value has not been recognised in the financial statements.

[2]

Q. 7) An investor cannot afford to construct a well-diversified portfolio. Which of the following best describes the significance of the beta of potential investments to that investor?

- A. The investor should focus on high beta stocks if the investor believes that the stock market will rise.
- B. Low beta investments will be a safer investment for the investor.
- C. The beta co-efficient is applicable only when the investor is actively investing in the market.
- D. Betas have no use for the investor.

[2]

Q. 8) An investment project has been evaluated using Monte Carlo simulation. After 10 lakh simulations, the results have stabilised and the expected net present value is positive and averages ₹1 crore, with a range of outcomes varying from minus ₹30 lakhs to plus ₹1.7 crore.

Which of the following statements best interprets these results?

- A. This is a low-risk project.
- B. The project satisfies the net present value criterion.
- C. The project fails the net present value criterion.
- D. The model should be run on more simulations.

[2]

Q. 9) Which of the following best describes a 'certainty equivalent' used in project appraisal?

- A. The expected value of an uncertain future cash inflow.
- B. The fixed amount that the decision maker would accept as an alternative to an uncertain expectation.
- C. The market price of an uncertain future cash flow calculated using different betas.
- D. The smallest amount that is anticipated from an uncertain future cash inflow.

[2]

Q. 10) A company owns a machinery that costs ₹10,000 on 1st December 2012. The equipment's estimated useful life was 10 years, and estimated scrap value was ₹1,000. What is the depreciation charge on this equipment for the year ended 30th November 2022, assuming the straight-line method of depreciation?

- A. ₹450
- B. ₹1,800
- C. ₹1,000
- D. ₹900

[2]

Q. 11) A road construction project requires a machinery which will have a useful lifetime of 3 years. The estimated cashflows are:

Time in Years	0	1	2	3
Cashflow in crores	(9)	2	2	6

(All revenues are received at the end of the year)

Calculate the following and comment on the results including any assumption used

- a) Annual capital charge
- b) Payback period
- c) Nominal returns over three years

[5]

Q. 12) Go-mobile is a listed company developing mobility aids. Go-mobile was purchased by a group of private shareholders and is no longer quoted on stock exchange. Discuss the advantages and disadvantages to various stakeholders for Go-mobile becoming a private company.

[5]

Q. 13) ABC is a multinational manufacturing company in the Cruise ship industry. A decision was taken at the board meeting to use derivatives instruments.

i) List the main type of derivative instruments that ABC can use. (2)

ii) Explain how these derivative instruments might be used by ABC. (8)

[10]

Q. 14) Manager of a large Pension fund came across an article on Covered Bonds in a magazine. The article said that Covered bonds existed in the country for almost a century and are also one of the highest issued bonds in the world. It is estimated that there are more than 1 lakh crores worth of covered bonds outstanding in the world. Manager is proposing to consider covered bonds as one of the investments for the pension fund and asked you to prepare a report on Covered bonds. Prepare a report explaining

i) What is a Covered Bond? (2)

ii) Advantages of investing in Covered Bonds. (4)

iii) Risks of investing in Covered bonds. (4)

[10]

Q. 15) The Chief Executive Officer AYZ company suggest a change in the capital structure of the company to finance its overall operations by borrowing rather than issuing further share capital. Describe how the capital structure of a company might affect the share price.

[6]

Q. 16) The external auditor has raised concerns that the main machine used in the manufacturing of toys should be depreciated more quickly than the company is currently depreciating it in its draft statements. The company's directors are not ready to revise their draft statements. What are the consequences for the company for not revising draft statements to address the auditors concern?

[4]

Q. 17) There are conflicting objectives between different stakeholders of any company. Give any one example of conflicts that may arise between

- i) Shareholders and Managers (1)
 - ii) Shareholders and providers of finance (1)
 - iii) Managers and workers (1)
 - iv) Shareholders and local community (1)
 - v) Managers and Customers (1)
- [5]**

Q. 18) A company's operating profit was ₹7,50,000. Depreciation of ₹2,77,000 was charged in arriving at this figure.

Comparing the balance sheet at the beginning and end of the period showed the following movements:

Inventories increase of ₹72,000

Trade receivables decrease of ₹20,000

Trade payables increase of ₹11,000

- i) Calculate the cash generated from operations for this company. (3)

This company owns a building worth ₹50 lakhs. The depreciation to date on the building is ₹2 lakhs. The directors of the company have decided to revalue the building to its fair value of ₹58 lakhs.

- ii) Calculate the balance on the company's revaluation reserve. (2)
- [5]**

Q. 19) XYZ Ltd is an Engineering Consultation company having two subsidiary companies named X and Y. These Subsidiary companies are operating in two different states and not in competition with each other.

The director of XYZ Ltd wants to compare the performance of the subsidiaries X and Y based on the latest annual reports. The annual reports are given as follows:

Income Statements for the year ending 31.03.2023 (In ₹ thousand)		
	<i>X</i>	<i>Y</i>
Revenue	1,900	3,100
Cost of sales	790	1,200
Gross profit	1,110	1,900
Advertising and distribution	310	603
Administration	221	182
Operating profit	579	1115
Interest	77	34
Net profit	502	1081

Statements of financial position as at 31.03.2023 (In ₹ thousand)		
	<i>X</i>	<i>Y</i>
Property, plant and equipment	3,200	3,700
Current assets		
Receivables	280	625
Total assets	3,480	4,325
Equity	2,200	3,831
Non-current liabilities		
Loans	920	415
Current liabilities		
Overdraft	360	79
Total equity and liabilities	3,480	4,325

i) Using relevant ratios, compare the performance of two subsidiaries in terms of profitability, liquidity and management of receivables. The comparison should include comments on the performance of X and Y based on the calculated ratios. (12)

ii) The director has decided to suggest some operational changes for poorly performing subsidiary. Explain the limitations of the analysis done in part (i) and what are the additional information the director may require to decide on the operational changes. (4)
[16]

Q. 20) Mint is a listed toy manufacturing company. The Director of the company wants a detailed report on the companies cost of capital. The following information about the company is given:

- The risk-free rate of interest is 6% per annum.
- The corporation tax rate is 22%.
- The ungeared beta on Mints's equity is 1.25.
- The debentures have a coupon rate of 5% and are redeemable at par in five years' time.
- The company has debentures with a face value of ₹320 Crores. Its market value is ₹260 Crores.
- The market capitalisation of the company's equity is ₹500 Crores.

- The market rate of return is 11% per annum.

Using the information given above:

- i) Determine Mint's cost of equity using the company's geared beta. (4)
- ii) Estimate Mint's approximate cost of debt. (4)
- iii) Calculate Mint's WACC. (2)

The director wants to expand the business either through borrowing or by issuing fresh equity. Many potential lenders are ready to offer a loan to Mint at a rate of 9.7%. Due to this the director is encouraged and feels that issuing of debt will reduce the company's WACC and make the expansion of the business more attractive in terms of Net Present Value (NPV).

- iv) Explain why the director is convinced that issuing debt rather than issuing equity will reduce the company's WACC and explain whether the director's understanding is correct. (4)
- [14]**
